

Dense Yes, Closed No for Algebras of Nowhere-Bounded Functions

Candidate full solution packet
arXiv:1207.3818, Problem 3

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Status. Candidate full answer in the source's intended cardinal/separable setting, pending expert review: $\mathcal{G}_p \cup \{0\}$ contains a dense free algebra, but it contains no nonzero L^p -closed subalgebra. The closed-algebra obstruction is valid on every measure space and for every $0 < p < \infty$.

1 Source problem and setting

Głab, Kaufmann, and Pellegrini define

$$\mathcal{G}_p = \{f \in L^p(\mu) : f \text{ is nowhere } L^\infty(\mu)\}$$

and ask the following on printed page 17 of [1]:

shows that g is nowhere essentially bounded. □

5.1. Comments and open problems. As a Corollary from Theorem 1.7(e) we have the following:

Corollary 5.1. *If μ is an atomless and outer regular positive Borel measure on X with full support and $0 < p < \infty$, then*

$$\mathcal{G}_p \doteq \{f \in L^p(\mu) : f \text{ is nowhere } L^\infty(\mu)\}$$

is strongly $\mathfrak{c}$ -algebrable.

It is a straightforward exercise for the reader to show, using a construction similar to the one used to prove Theorem 1.7(a), that $\mathcal{G}_p$ is spaceable in L^p . To finish this section we pose the following problem:

Problem 3. Does $\mathcal{G}_p \cup \{0\}$ admit *dense* or *closed* subalgebras of L^p ?

6. WHEN ARE THERE NOWHERE q -INTEGRABLE FUNCTIONS IN L^p ?

We conclude this work with a couple of remarks and questions on necessary/sufficient conditions on a positive Borel measure space (X, μ) , so that there exist nowhere q -integrable

The positive part below uses exactly the factorial dense partition constructed in the proof of Theorem 1.7(e) of [1]. Namely, there are pairwise disjoint Borel sets $(B_j)_{j \geq 1}$ such that

$$\mu(B_j) \leq \frac{1}{j!}, \quad \mu(U \cap B_j) > 0 \tag{1}$$

for every j and every nonempty open $U \subset X$. The source obtains these sets when X has a countable π -base and μ is atomless, outer regular, and has full support.

For density, we make explicit the same cardinal convention used in the source's dense-lineability proof:

$$\text{dens } L^p(\mu) \leq \mathfrak{c}. \quad (2)$$

This holds in particular whenever $L^p(\mu)$ is separable, including $[0, 1]$ with Lebesgue measure and the standard examples highlighted in the paper. The source itself invokes separability/cardinality $\mathfrak{c}$ in its proof of Theorem 1.7(d).

2 Proof intuition

The source already provides free generators

$$g_\theta = \sum_{j \geq 1} \theta^j \chi_{B_j}, \quad \theta > 1.$$

Factorial decay makes every polynomial in them p -integrable, while each open set sees all the exponentially growing levels. Their algebra is not dense because these functions are constant on every B_j . To repair this, add bounded L^p functions chosen from a dense family and make the pathological perturbations arbitrarily small in L^p . Rational independence of the logarithms of the bases gives every nonzero polynomial a unique largest exponential rate. Bounded perturbations only create smaller rates, so this leading term cannot cancel on B_j .

The negative direction is more general. In a closed subalgebra $A \subset L^p$, fix $f \in A$. A Baire argument shows that multiplication by f is a bounded operator on A , even for $p < 1$. Applying its norm estimate to the powers f^n gives a uniform bound on $\|f\|_{np}$ as $n \rightarrow \infty$, which forces $f \in L^\infty$. Thus every closed L^p subalgebra consists entirely of bounded functions and cannot have a nonzero member of $\mathcal{G}_p$.

3 The dense free algebra

Theorem 1. *Let $0 < p < \infty$. Assume (1) and (2). Then $\mathcal{G}_p \cup \{0\}$ contains a dense free real algebra in $L^p(\mu)$.*

Proof. Bounded L^p functions are dense in L^p : truncate any representative in absolute value. Choose a dense family

$$D \subset L^p(\mu) \cap L^\infty(\mu), \quad |D| \leq \mathfrak{c}.$$

Index a family by $I = D \times \mathbb{N}$. Since $|I| \leq \mathfrak{c}$, choose numbers $(\theta_i)_{i \in I} \subset (1, \infty)$ such that $(\log \theta_i)_{i \in I}$ is linearly independent over $\mathbb{Q}$. Put

$$g_i = \sum_{j=1}^{\infty} \theta_i^j \chi_{B_j}.$$

By (1),

$$\|g_i\|_p^p \leq \sum_{j=1}^{\infty} \frac{\theta_i^{pj}}{j!} < \infty.$$

Write $i = (h, n)$ with $h \in D$ and $n \in \mathbb{N}$, choose $\varepsilon_i > 0$ so small that

$$\|\varepsilon_i g_i\|_p < \frac{1}{n},$$

and define

$$z_i = h + \varepsilon_i g_i.$$

For every fixed $h \in D$, the sequence $(z_{(h,n)})_n$ converges to h in L^p . Hence the set of generators $(z_i)_{i \in I}$, and therefore the algebra it generates, is dense in L^p .

We prove that every nonzero polynomial without constant term in finitely many distinct generators belongs to $\mathcal{G}_p$. Let

$$P(t_1, \dots, t_r) = \sum_{\alpha \in F} c_\alpha t^\alpha, \quad F \subset \mathbb{N}_0^r \setminus \{0\},$$

where all $c_\alpha \neq 0$. For the selected generators set

$$\lambda_\alpha = \prod_{\ell=1}^r \theta_{i_\ell}^{\alpha_\ell}.$$

Rational independence of the logarithms makes the λ_α pairwise distinct. Let α^* be the unique index with largest λ_α .

On B_j we have

$$z_{i_\ell}(x) = h_\ell(x) + \varepsilon_\ell \theta_{i_\ell}^j, \quad \|h_\ell\|_\infty < \infty.$$

Expanding the polynomial, its unique term with the largest exponential rate is

$$c_{\alpha^*} \varepsilon^{\alpha^*} \lambda_{\alpha^*}^j, \quad \varepsilon^{\alpha^*} := \prod_{\ell=1}^r \varepsilon_\ell^{\alpha_\ell^*}.$$

Every other expanded term has a rate strictly smaller than λ_{α^*} and a coefficient uniformly bounded in x . Since only finitely many terms occur, there are $C > 0$ and $1 \leq \rho < \lambda_{\alpha^*}$ such that, uniformly for $x \in B_j$,

$$\left| P(z_{i_1}, \dots, z_{i_r})(x) - c_{\alpha^*} \varepsilon^{\alpha^*} \lambda_{\alpha^*}^j \right| \leq C \rho^j. \quad (3)$$

It follows that for all sufficiently large j ,

$$|P(z_{i_1}, \dots, z_{i_r})(x)| \geq \frac{1}{2} |c_{\alpha^*}| \varepsilon^{\alpha^*} \lambda_{\alpha^*}^j \quad (x \in B_j). \quad (4)$$

For completeness, the polynomial is in L^p . On B_j its absolute value is at most $C'(1 + \Lambda^j)$ for fixed C', Λ , so (1) and factorial decay make the p th-power integral over $\bigcup_j B_j$ finite. Off this union all g_i vanish. There the polynomial is a sum of positive-degree monomials in bounded L^p functions, and every such monomial is in L^p . This argument does not use convexity and is valid for $p < 1$ as well.

Finally, every nonempty open U meets every B_j in positive measure. The lower bound (4) tends to infinity, so the polynomial is not essentially bounded on U . It is therefore in $\mathcal{G}_p$. In particular no nonzero polynomial relation exists, proving that the generators are free. $\square$

4 Why a closed algebra is impossible

Theorem 2. *Let (X, Σ, μ) be any measure space and $0 < p < \infty$. If A is a pointwise subalgebra of $L^p(\mu)$ which is closed in the L^p topology, then*

$$A \subset L^\infty(\mu).$$

Consequently $\mathcal{G}_p \cup \{0\}$ contains no nonzero closed subalgebra.

Proof. Fix $f \in A$. For $m \in \mathbb{N}$, let

$$E_m = \{g \in A : \|fg\|_p \leq m\}.$$

Each E_m is closed in A . Indeed, if $g_k \rightarrow g$ in L^p , select a subsequence converging almost everywhere and use Fatou's lemma to obtain

$$\|fg\|_p^p \leq \liminf_k \|fg_k\|_p^p.$$

Moreover $A = \bigcup_m E_m$, because A is an algebra and hence $fg \in A \subset L^p$ for every $g \in A$.

The closed subspace A is a complete metrizable topological vector space, also when $p < 1$. By Baire's theorem, some E_m has nonempty interior. Thus there are $g_0 \in A$ and $r > 0$ such that

$$g_0 + \{h \in A : \|h\|_p < r\} \subset E_m.$$

Since also $g_0 \in E_m$, the triangle inequality when $p \geq 1$, and the p -triangle inequality when $p < 1$, give a constant K_p such that

$$\|fh\|_p \leq K_p m \quad (\|h\|_p < r).$$

Scaling now yields a constant $C_f < \infty$ with

$$\|fh\|_p \leq C_f \|h\|_p \quad (h \in A). \tag{5}$$

Apply (5) repeatedly to $h = f^{n-1}$. Since all powers belong to A ,

$$\|f^n\|_p \leq C_f^{n-1} \|f\|_p, \quad \text{so} \quad \|f\|_{np} \leq C_f^{(n-1)/n} \|f\|_p^{1/n}.$$

We claim that $\|f\|_\infty \leq C_f$. Otherwise choose $t > C_f$ such that $E = \{|f| > t\}$ has positive measure. Since $f \in L^p$, this measure is also finite. Therefore

$$\|f\|_{np} \geq t \mu(E)^{1/(np)} \rightarrow t,$$

contradicting (6). Thus $f \in L^\infty$. As $f \in A$ was arbitrary, $A \subset L^\infty$.

If in addition $A \subset \mathcal{G}_p \cup \{0\}$, any nonzero $f \in A$ would be globally essentially bounded and nowhere essentially bounded simultaneously, an impossibility. Hence $A = \{0\}$. $\square$

5 Conclusion, scope, and novelty

Combining Theorems 1 and 2 gives the sharp dichotomy requested in Problem 3:

dense free algebra: yes; nonzero closed algebra: no.
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The dense statement applies under the factorial partition used by the source and the cardinal condition (2), in particular throughout the separable setting explicitly used in the paper and for $L^p[0, 1]$. If the source's displayed topological hypotheses are interpreted to include exotic L^p spaces of density strictly above $\mathfrak{c}$, no claim about dense algebras in that additional cardinal regime is made. The negative closed-algebra theorem is unrestricted.

The proof was audited separately at the unique-leading-rate step, the off-partition integrability step, the $p < 1$ Baire estimate, and the passage from power bounds to essential boundedness. The accompanying `verification.md` records these checks; the finite script is only a sanity check and is not used as proof.

A bounded novelty search on 11 August 2026 covered the four cheap run indexes, the locally parsed arXiv corpus, and current arXiv-domain searches for the exact title, arXiv id, Problem 3 wording, “nowhere essentially bounded”, “dense algebra”, and “closed subalgebra”. It recovered the source and its predecessor arXiv:1204.6404, which state the question as open, but no later answer. Novelty remains subject to specialist review.

Human-review recommendation. Review as a candidate full answer in the paper’s intended setting. Check especially that condition (2) matches the source’s standing cardinal convention; independently, Theorem 2 is a fully general negative theorem.

References

- [1] S. Głab, P. L. Kaufmann, and L. Pellegrini, *Large structures made of nowhere L^p functions*, Studia Math. 216 (2013), no. 1, 51–69; arXiv:1207.3818, Problem 3 on printed page 17.